Problem Set 2 (Analytical Part)

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Question 1

The complete-data likelihood is:

$$\mathcal{L}_c(\theta) = p(\boldsymbol{y} \mid \theta) = \frac{n!}{y_1!y_2!y_3!y_4!y_5!} \prod_{k=1}^{5} \delta_k^{y_k}$$

The complete-data log-likelihood is:

$$\begin{aligned}
\ell_c(\theta) &= \log n! - \sum_{k=1}^{5} \log y_k! + \sum_{k=1}^{5} y_k \log \delta_k \\
&= y_1 \log\left(\frac{1}{4}\theta\right) + y_2 \log\left(\frac{1}{4}(1-\theta)\right) + y_3 \log\left(\frac{1}{4}(1-\theta)\right) + y_4 \log\left(\frac{1}{2}\right) + y_5 \log\left(\frac{1}{4}\theta\right) + C \\
&= y_1 \log\theta + y_2 \log(1-\theta) + y_3 \log(1-\theta) + y_5\theta + C \\
&= (y_1 + y_5)\log\theta + (y_2 + y_3)\log(1-\theta) + C
\end{aligned}$$

Since we have only observed $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ where $x_4 = y_4 + y_5$ and $y_1 = x_1$, $y_2 = x_2$, $y_3 = x_3$ given the setup of the question, the expected log-likelihood is:

$$\mathbb{E}[\ell_c(\theta) \mid \boldsymbol{X} = \boldsymbol{x}] = \log\theta(x_1 + \mathbb{E}[y_5 \mid \boldsymbol{X} = \boldsymbol{x}]) + (x_2 + x_3)\log(1-\theta) + C$$

Therefore, we need to derive the conditional expectation of y_5 for the EM algorithm.
Let $Y_5 = k$. Note that:

$$\Pr(Y_5 = k \mid \boldsymbol{X} = \boldsymbol{x}) = \frac{\Pr(Y_5 = k, \boldsymbol{X} = \boldsymbol{x})}{\Pr(\boldsymbol{X} = \boldsymbol{x})}$$

Since $x_4 = y_4 + y_5$, we have $y_4 = x_4 - k$. Therefore, given that $Y_1 = x_1$, $Y_2 = x_2$, $Y_3 = x_3$, $Y_4 = x_4 - k$ and $Y_5 = k$ and $\boldsymbol{Y} \sim \text{Mult}(n, \delta_1, \delta_2, \delta_3, \delta_4, \delta_5)$, we have:

$$\Pr(Y_5 = k, \boldsymbol{X} = \boldsymbol{x}) = \frac{n!}{x_1!x_2!x_3!(x_4-k)!k!} \delta_1^{x_1}\delta_2^{x_2}\delta_3^{x_3}\delta_4^{x_4-k}\delta_5^{k}$$

Since $\boldsymbol{X} \sim \text{Mult}(n, \pi_1, \pi_2, \pi_3, \pi_4)$ and $\pi_1 = \delta_1$, $\pi_2 = \delta_2$, $\pi_3 = \delta_3$, and $\pi_4 = \delta_4 + \delta_5$. We have:

$$\Pr(\boldsymbol{X} = \boldsymbol{x}) = \frac{n!}{x_1!x_2!x_3!x_4!} \delta_1^{x_1}\delta_2^{x_2}\delta_3^{x_3}(\delta_4 + \delta_5)^{x_4}$$

Therefore, we have:

$$\begin{aligned}\Pr(Y_5 = k \mid \boldsymbol{X} = \boldsymbol{x}) &= \frac{x_4!}{(x_4-k)!k!}\delta_4^{x_4-k}\delta_5^k \times (\delta_4+\delta_5)^{x_4} \\ &= \binom{x_4}{k}\left(\frac{\delta_4}{\delta_4+\delta_5}\right)^{x_4-k}\left(\frac{\delta_5}{\delta_4+\delta_5}\right)^{k} \\ &= \binom{x_4}{k}\left(\frac{\delta_4}{\delta_4+\delta_5}\right)^{x_4-k}\left(1-\frac{\delta_4}{\delta_4+\delta_5}\right)^{k}\end{aligned}$$

Note that:

$$\frac{\delta_4}{\delta_4+\delta_5} = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{4}\theta} = \frac{2}{2+\theta}$$

Therefore, y_5 conditional on $\boldsymbol{x}$ follows a binomial distribution:

$$y_5 \mid \boldsymbol{X} = \boldsymbol{x} \sim \textbf{Bin}\left(x_4,\, \frac{2}{2+\theta}\right)$$

Therefore,

$$\mathbb{E}[y_5 \mid \boldsymbol{X} = \boldsymbol{x}] = x_4 \times \frac{2}{2+\theta} = \frac{2x_4}{2+\theta}$$

- **E-Step:** Assume the current parameter estimate for θ is $\theta^{(t)}$. Then the expected complete-data log-likelihood is:

$$Q\left(\theta \mid \theta^{(t)}\right) = \mathbb{E}\left[\ell(\theta) \mid \boldsymbol{X} = \boldsymbol{x}, \theta^{(t)}\right] = \log\theta\left(x_1 + \frac{2x_4}{2+\theta^{(t)}}\right) + (x_2+x_3)\log(1-\theta) + C$$

- **M-Step:** We maximize the Q-function. We examine the first-order condition:

$$\frac{\partial Q(\theta \mid \theta^{(t)})}{\partial\theta} = \frac{1}{\theta}\left(x_1 + \frac{2x_4}{2+\theta^{(t)}}\right) - (x_2+x_3)\frac{1}{1-\theta} = 0$$

  Solving for θ, we get:

$$\theta^{(t+1)} = \frac{x_1 + x_4\frac{\theta^{(t)}}{2+\theta^{(t)}}}{x_1+x_2+x_3+x_4\frac{\theta^{(t)}}{2+\theta^{(t)}}}$$

Question 2

Assume that M_{ij} follows a Bernoulli distribution:

$$M_{ij} \sim \textbf{Bernoulli}(p)$$

where

$$p = \Pr(M_{ij} = 1)$$

For each variable k, define:

$$\begin{cases} m_k = \Pr(\delta_k(i,j) = 1 \mid M_{ij} = 1) \\ u_k = \Pr(\delta_k(i,j) = 1 \mid M_{ij} = 0) \end{cases}$$

Therefore

$$\delta_k(i,j) \mid M_{ij} = 1 \sim \textbf{Bernoulli}(m_k), \quad \delta_k(i,j) \mid M_{ij} = 0 \sim \textbf{Bernoulli}(u_k)$$

Conditional on $M_{ij} = 1$, the components are independent

$$\Pr(\boldsymbol{\delta}(i,j) \,|\, M_{ij} = 1) = \prod_{k=1}^{K} \Pr(\delta_k(i,j) \,|\, M_{ij} = 1) = \prod_{k=1}^{K} m_k^{\delta_k(i,j)} (1 - m_k)^{1-\delta_k(i,j)}$$

Similarly, we have:

$$\Pr(\boldsymbol{\delta}(i,j) \,|\, M_{ij} = 0) = \prod_{k=1}^{K} u_k^{\delta_k(i,j)} (1 - u_k)^{1-\delta_k(i,j)}$$

Therefore, the marginal distribution of the observed $\boldsymbol{\delta}(i,j)$ is:

$$\Pr(\boldsymbol{\delta}(i,j)) = p \prod_{k=1}^{K} m_k^{\delta_k(i,j)} (1 - m_k)^{1-\delta_k(i,j)} + (1-p) \prod_{k=1}^{K} u_k^{\delta_k(i,j)} (1 - u_k)^{1-\delta_k(i,j)}$$

For notational convenience, we denote each record pair (i,j) by r, $r = 1, 2, \cdots, n$. Therefore

$$\Pr(\boldsymbol{\delta}_r) = p \prod_{k=1}^{K} m_k^{\delta_{rk}} (1 - m_k)^{1-\delta_{rk}} + (1-p) \prod_{k=1}^{K} u_k^{\delta_{rk}} (1 - u_k)^{1-\delta_{rk}}$$

For one record pair r, the complete-data likelihood is:

$$\begin{aligned}
\Pr(M_r, \boldsymbol{\delta}_r) &= \Pr(M_r) \Pr(\boldsymbol{\delta}_r \,|\, M_r) \\
&= \left(\Pr(M_r = 1) \Pr(\boldsymbol{\delta}_r \,|\, M_r = 1)\right)^{M_r} \left(\Pr(M_r = 0) \Pr(\boldsymbol{\delta}_r \,|\, M_r = 0)\right)^{1-M_r} \\
&= \left(p \prod_{k=1}^{K} m_k^{\delta_{rk}} (1 - m_k)^{1-\delta_{rk}} \right)^{M_r} \left((1-p) \prod_{k=1}^{K} u_k^{\delta_{rk}} (1 - u_k)^{1-\delta_{rk}} \right)^{1-M_r}
\end{aligned}$$

For n record pairs, the complete-data log-likelihood w.r.t. $\boldsymbol{\theta} = (p, \{m_k, u_k\})$ is:

$$\begin{aligned}
\ell_c(\boldsymbol{\theta}) &= \sum_{r=1}^{n} \log \Pr(M_r, \boldsymbol{\delta}_r) \\
&= \sum_{r=1}^{n} M_r \log \left(p \prod_{k=1}^{K} m_k^{\delta_{rk}} (1 - m_k)^{1-\delta_{rk}} \right) + (1 - M_r) \log \left((1-p) \prod_{k=1}^{K} u_k^{\delta_{rk}} (1 - u_k)^{1-\delta_{rk}} \right) \\
&= \sum_{r=1}^{n} M_r \left[\log p + \sum_{k=1}^{K} \left(\delta_{rk} \log m_k + (1 - \delta_{rk}) \log(1 - m_{rk}) \right) \right] \\
&+ \sum_{r=1}^{n} (1 - M_r) \left[\log(1-p) + \sum_{k=1}^{K} \left(\delta_{rk} \log u_k + (1 - \delta_{rk}) \log(1 - u_{rk}) \right) \right]
\end{aligned}$$

- **E-Step**:
  As shown above, given the parameters, we need to derive the conditional expectation of M_r because it is unobservable.
  Suppose we have current parameter estimates:

  $$\boldsymbol{\theta}^{(t)} = \left(p^{(t)}, m_k^{(t)}, u_k^{(t)} \right)$$

  Since M_r is a binary variable, we have:

  $$\mathbb{E}\left[M_r \,|\, \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)} \right] = \Pr\left(M_r = 1 \,|\, \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)} \right)$$

Define:

$$\gamma_r^{(t)} = \mathbb{E}\left[M_r \mid \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)}\right] = \Pr\left(M_r = 1 \mid \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)}\right)$$

By Bayes' rule,

$$\begin{aligned}
\gamma_r^{(t)} = \Pr\left(M_r = 1 \mid \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)}\right) &= \frac{\Pr(M_r = 1 \mid \boldsymbol{\theta}^{(t)}) \Pr(\boldsymbol{\delta}_r \mid M_r = 1, \boldsymbol{\theta}^{(t)})}{\Pr(\boldsymbol{\delta}_r \mid \boldsymbol{\theta}^{(t)})} \\
&= \frac{p^{(t)} \prod_{k=1}^{K} \left(m_k^{(t)}\right)^{\delta_{rk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{rk}}}{p^{(t)} \prod_{k=1}^{K} \left(m_k^{(t)}\right)^{\delta_{rk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{rk}} + (1 - p^{(t)}) \prod_{k=1}^{K} \left(u_k^{(t)}\right)^{\delta_{rk}} \left(1 - u_k^{(t)}\right)^{1-\delta_{rk}}}
\end{aligned}$$

The Q-function is:

$$\begin{aligned}
Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) &= \mathbb{E}\left[\ell_c(\boldsymbol{\theta}) \mid \boldsymbol{\delta}, \boldsymbol{\theta}^{(t)}\right] \\
&= \sum_{r=1}^{n} \mathbb{E}\left[M_r \mid \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)}\right] \left[\log p + \sum_{k=1}^{K} \left(\delta_{rk} \log m_k + (1 - \delta_{rk}) \log(1 - m_{rk})\right)\right] \\
&+ \sum_{r=1}^{n} \left(1 - \mathbb{E}\left[M_r \mid \boldsymbol{\delta}_r, \boldsymbol{\theta}^{(t)}\right]\right) \left[\log(1 - p) + \sum_{k=1}^{K} \left(\delta_{rk} \log u_k + (1 - \delta_{rk}) \log(1 - u_{rk})\right)\right] \\
&= \sum_{r=1}^{n} \gamma_r^{(t)} \left[\log p + \sum_{k=1}^{K} \left(\delta_{rk} \log m_k + (1 - \delta_{rk}) \log(1 - m_{rk})\right)\right] \\
&+ \sum_{r=1}^{n} \left(1 - \gamma_r^{(t)}\right) \left[\log(1 - p) + \sum_{k=1}^{K} \left(\delta_{rk} \log u_k + (1 - \delta_{rk}) \log(1 - u_{rk})\right)\right]
\end{aligned}$$

- **M-Step**: Solving for p:

$$\frac{\partial Q}{\partial p} = \frac{1}{p} \sum_{r=1}^{n} \gamma_r^{(t)} - \frac{1}{1 - p} \sum_{r=1}^{n} \left(1 - \gamma_r^{(t)}\right) = 0$$

This gives:

$$p^{(t+1)} = \frac{1}{n} \sum_{r=1}^{n} \gamma_r^{(t)}$$

Solving for m_k:

$$\frac{\partial Q}{\partial m_k} = \sum_{r=1}^{n} \gamma_r^{(t)} \left(\delta_{rk} \times \frac{1}{m_k} - (1 - \delta_{rk}) \times \frac{1}{1 - m_k}\right) = 0$$

This gives:

$$m_k^{(t+1)} = \frac{\sum_{r=1}^{n} \gamma_r^{(t)} \delta_{rk}}{\sum_{r=1}^{n} \gamma_r^{(t)}}$$

Since the component of the Q-function with u_k has the same structure as that with m_k, solving $\frac{\partial Q}{\partial u_k} = 0$ for u_k gives:

$$u_k^{(t+1)} = \frac{\sum_{r=1}^{n} \left(1 - \gamma_r^{(t)}\right) \delta_{rk}}{\sum_{r=1}^{n} 1 - \gamma_r^{(t)}}$$

Question 3

Given the specific structure of the data, we need to tweak the parameter estimators derived for Question 2.

We need to estimate the parameter for the latent variable M_{ij} and probability parameters for each δ:

$$\boldsymbol{\theta} = (p, m_1, m_2, m_3, u_1, u_2, u_3)$$

Let g denote an agreement pattern, $g = 1, \cdots, 5$. For each pattern:

$$\boldsymbol{\delta}_g = (\delta_{g1}, \cdots, \delta_{g5})$$

Let G_g denote the set of record pairs under g:

$$G_g = \{r : \boldsymbol{\delta}_r = \boldsymbol{\delta}_g\}$$

Apparently, for record pairs with the same agreement pattern, $\boldsymbol{\delta}$ is the same, hence γ being the same as well. Therefore, for each $r \in G_g$:

$$\gamma_r^{(t)} = \gamma_g^{(t)} = \frac{p^{(t)} \prod_{k=1}^{3} \left(m_k^{(t)}\right)^{\delta_{gk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{gk}}}{p^{(t)} \prod_{k=1}^{3} \left(m_k^{(t)}\right)^{\delta_{gk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{gk}} + (1 - p^{(t)}) \prod_{k=1}^{3} \left(u_k^{(t)}\right)^{\delta_{gk}} \left(1 - u_k^{(t)}\right)^{1-\delta_{gk}}}$$

For notational convenience, we define two functions named $\delta - m$ and $\delta - u$ separately:

$$f(\boldsymbol{\delta}_g, \boldsymbol{m}^{(t)}) = \prod_{k=1}^{3} \left(m_k^{(t)}\right)^{\delta_{gk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{gk}}$$

$$f(\boldsymbol{\delta}_g, \boldsymbol{u}^{(t)}) = \prod_{k=1}^{3} \left(u_k^{(t)}\right)^{\delta_{gk}} \left(1 - u_k^{(t)}\right)^{1-\delta_{gk}}$$

$$\log f(\boldsymbol{\delta}_g, \boldsymbol{m}^{(t)}) = \sum_{k=1}^{3} \delta_{gk} \log \left(m_k^{(t)}\right) + \left(1 - \delta_{gk}\right) \log \left(1 - m_k^{(t)}\right)$$

$$\log f(\boldsymbol{\delta}_g, \boldsymbol{u}^{(t)}) = \sum_{k=1}^{3} \delta_{gk} \log \left(u_k^{(t)}\right) + \left(1 - \delta_{gk}\right) \log \left(1 - u_k^{(t)}\right)$$

where

$$\boldsymbol{m}^{(t)} = (m_1^{(t)}, m_2^{(t)}, m_3^{(t)})$$

$$\boldsymbol{u}^{(t)} = (u_1^{(t)}, u_2^{(t)}, u_3^{(t)})$$

Therefore, the responsibility variable γ at iteration t under pattern g is:

$$\gamma_g^{(t)} = \frac{p^{(t)} f(\boldsymbol{\delta}_g, \boldsymbol{m}^{(t)})}{p^{(t)} f(\boldsymbol{\delta}_g, \boldsymbol{m}^{(t)}) + (1 - p^{(t)}) f(\boldsymbol{\delta}_g, p^{(t)}, \boldsymbol{u}^{(t)})}$$

For each iteration t, the observed-data log-likelihood under an agreement pattern g for one observation is:

$$\ell_o(\boldsymbol{\theta}^{(t)}; g) = \log \Pr(\boldsymbol{\delta}_g \mid \boldsymbol{\theta}^{(t)})$$

For each iteration t, the observed-data log-likelihood for all observations is:

$$\ell_o(\boldsymbol{\theta}^{(t)}) = \sum_{r=1}^{n} \log \Pr(\boldsymbol{\delta}_r \mid \boldsymbol{\theta}^{(t)}) = \sum_{g=1}^{5} \sum_{r \in G_g} \log \Pr(\boldsymbol{\delta}_r \mid \boldsymbol{\theta}^{(t)}) = \sum_{g=1}^{5} |G_g| \ell_o(\boldsymbol{\theta}^{(t)}; g)$$

where

$$\begin{aligned}
\Pr(\boldsymbol{\delta}_g \mid \boldsymbol{\theta}^{(t)}) &= p^{(t)} \prod_{k=1}^{3} \left(m_k^{(t)}\right)^{\delta_{gk}} \left(1 - m_k^{(t)}\right)^{1-\delta_{gk}} + (1 - p^{(t)}) \prod_{k=1}^{K} \left(u_k^{(t)}\right)^{\delta_{gk}} \left(1 - u_k^{(t)}\right)^{1-\delta_{gk}} \\
&= p^{(t)} f(\boldsymbol{\delta}_g, \boldsymbol{m}^{(t)}) + (1 - p^{(t)}) f(\boldsymbol{\delta}_g, \boldsymbol{u}^{(t)})
\end{aligned}$$

Correspondingly, the parameter update for p is:

$$p^{(t+1)} = \frac{1}{n} \sum_{r=1}^{n} \gamma_r^{(t)} = \frac{1}{n} \sum_{g=1}^{5} \sum_{r \in G_g} \gamma_r^{(t)} = \frac{1}{n} \sum_{g=1}^{5} |G_g| \gamma_g^{(t)}$$

Similarly, for m_k and u_k, we have:

$$m_k^{(t+1)} = \frac{\sum_{g=1}^{5} \gamma_g^{(t)} |G_g| \delta_{gk}}{\sum_{g=1}^{5} |G_g| \gamma_g^{(t)}}$$

$$u_k^{(t+1)} = \frac{\sum_{g=1}^{5} \left(1 - \gamma_g^{(t)}\right) |G_g| \delta_{gk}}{\sum_{g=1}^{5} |G_g| \left(1 - \gamma_g^{(t)}\right)}$$

Now we are ready to implement the EM algorithm in `R`. The results are presented in the file named "em_q3_record_linkage.pdf."

Question 4

Suppose $X_1, \cdots, X_n$ are independent observations generated from a two-component Gaussian mixture:

$$X_i \sim \pi_1 \mathcal{N}(\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(\mu_2, \sigma_2^2)$$

where

$$\pi_1 + \pi_2 = 1, \pi_k > 0$$

Introduce an component indicator $Z_i \in \{1, 2\}$, with:

$$P(Z_i = k) = \pi_k, \quad k = 1, 2$$

Conditional on component membership:

$$X_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \sigma_k 2)$$

The parameters to be estimated are:

$$\boldsymbol{\theta} = (\pi_1, \pi_2, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$$

Define the membership indicator:

$$\delta_{ik} = \mathbb{I}(Z_i = k), \quad k = 1, 2, \quad \text{where } \delta_{i1} + \delta_{i2} = 1$$

Let $f_k(X_i)$ denote the Gaussian density for component k. If every component label Z_i were observed, the complete-data log-likelihood would be:

$$\begin{aligned}
\ell_c(\boldsymbol{\theta}) &= \log \prod_{i=1}^{n} \prod_{k=1}^{2} [\pi_k f_k(X_i)]^{\delta_{ik}} \\
&= \sum_{i=1}^{2} \sum_{k=1}^{2} \delta_{ik} \left[\log \pi_k + \log f_k(X_i)\right] \\
&= \sum_{i=1}^{2} \sum_{k=1}^{2} \delta_{ik} \left[\log \pi_k - \frac{1}{2} \log(2\pi\sigma_k^2) - \frac{(X_i - \mu_k)^2}{2\sigma_k^2}\right]
\end{aligned}$$

- **E-Step**:
  At iteration t, suppose our current parameter estimates are

  $$\boldsymbol{\theta}^{(t)} = \left(\pi_k^{(t)}, \mu_k^{(t)}, \sigma_k^{2(t)}\right)$$

  We construct the Q-function:

  $$Q\left(\boldsymbol{\theta} \,|\, \boldsymbol{\theta}^{(t)}\right) = \mathbb{E}_{\boldsymbol{\theta}^{(t)}} \left[\ell_c(\boldsymbol{\theta} \,|\, \text{ observed data }]\right.$$

  We only need to compute the responsibility variable:

  $$\gamma_{ik}^{(t)} = \mathbb{E}_{\theta^{(t)}} \left[\delta_{ik} \,|\, \text{observed data}\right]$$

  1. **Case 1: Observed class label** $(i \leq n_1)$
     Because Z_i is known, there is no uncertainty about the component membership.

     $$\gamma_{ik}^{(t)} = \mathbb{I}(Z_i = k), \quad i = 1, \cdots, n_1$$

     $$\begin{cases} Z_i = 1 \implies (\gamma_{i1}, \gamma_{i2}) = (1, 0) \\ Z_i = 2 \implies (\gamma_{i1}, \gamma_{i2}) = (0, 1) \end{cases}$$

  2. **Case 2: Missing class label** $(n_1 < i \leq n)$
     For the unlabeled observations, we derive the posterior probability of belonging to k:

     $$\begin{aligned}
     \gamma_{ik}^{(t)} &= P(Z_i = k \,|\, X_i, \boldsymbol{\theta}^{(t)}) = \frac{P(X_i \,|\, Z_i = k, \boldsymbol{\theta}^{(t)}) P(Z_i = k \,|\, \boldsymbol{\theta}^{(t)})}{P(X_i \,|\, \boldsymbol{\theta}^{(t)})} \\
     &= \frac{\pi_k^{(t)} f_k(X_i; \mu_k^{(t)}, \sigma_k^{2(t)})}{\sum_{j=1}^{2} \pi_j^{(t)} f_j(X_i; \mu_j^{(t)}, \sigma_j^{2(t)})}
     \end{aligned}$$

As such, the Q-function is:

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) \sum_{i=1}^{2} \sum_{k=1}^{2} \gamma_{ik}^{(t)} \left[\log \pi_k - \frac{1}{2} \log(2\pi\sigma_k^2) - \frac{(X_i - \mu_k)^2}{2\sigma_k^2} \right]$$

- **M-Step**:
  Define the effective number of observations assigned to k:

$$N_k^{(t)} = \sum_{i=1}^{n} \gamma_{ik}^{(t)}$$

To update the mixing proportions π_k, we only need to maximize:

$$\pi_k^{(t+1)} = \text{argmax}_{\pi_k} \sum_{i=1}^{n} \sum_{k=1}^{2} \gamma_{ik}^{(t)} \log \pi_k = \sum_{k=1}^{2} N_k^{(t)} \log \pi_k \text{ s.t. } \sum_{k=1}^{2} \pi_k = 1$$

Using a Lagrange multiplier λ, we have:

$$\mathcal{L} = \sum_{k=1}^{2} N_k^{(t)} \log \pi_k + \lambda \left(1 - \sum_{k=1}^{2} \pi_k \right)$$

Examine the FOC:

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{N_k^{(t)}}{\pi_k} - \lambda = 0 \implies \pi_k = \frac{N_k^{(t)}}{\lambda}$$

Since $\sum_{k=1}^{2} \pi_k = 1$,

$$\begin{aligned} 1 &= \sum_{k=1}^{2} \frac{N_k^{(t)}}{\lambda} = \sum_{k=1}^{2} \frac{1}{\lambda} \sum_{i=1}^{n} \gamma_{ik}^{(t)} \\ &= \frac{1}{\lambda} \sum_{i=1}^{n} \sum_{k=1}^{2} \gamma_{ik}^{(t)} = \frac{n}{\lambda} \end{aligned}$$

Therefore, $\lambda = n$. As such, we have:

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} \gamma_{ik}^{(t)}$$

For μ_k:

$$\mu_k^{(t+1)} = \text{argmax}_{\mu_k} - \frac{1}{2\sigma_k^2} \sum_{i=1}^{n} \gamma_{ik}^{(t)} (X_i - \mu_k)^2$$

Solving for μ_k:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{ik}^{(t)} X_i}{\sum_{i=1}^{n} \gamma_{ik}^{(t)}}$$

For σ_k^2:

$$\sigma_k^{2(t+1)} = -\frac{1}{2} \sum_{i=1}^{n} \gamma_{ik}^{(t)} \left[\log(2\pi\sigma_k^2) + \frac{(X_i - \mu_k)^2}{\sigma_k^2} \right]$$

Solving for σ_k^2:

$$\sigma_k^{2(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{ik}^{(t)} \left(X_i - \mu_k^{(t+1)}\right)^2}{\sum_{i=1}^{n} \gamma_{ik}^{(t)}}$$

Let $L_k = \{i \leq n : Z_i = k\}$ and $n_{1k} = |L_k|$.

$$\mu_k^{(t+1)} = \frac{\sum\limits_{i \in L_k} X_i + \sum\limits_{i=n_1+1}^{n} \gamma_{ik}^{(t)} X_i}{n_{1k} + \sum\limits_{i=n_1+1}^{n} \gamma_{ik}^{(t)}}$$

$$\pi_k^{(t+1)} = \frac{n_{1k} + \sum\limits_{i=n_1+1}^{n} \gamma_{ik}^{(t)}}{n}$$

$$\sigma_k^{2(t+1)} = \frac{\sum\limits_{i \in L_k} (X_i - \mu_k^{(t+1)})^2 + \sum\limits_{i=n_1+1}^{n} \gamma_{ik}^{(t)} (X_i - \mu_k^{(t+1)})^2}{n_{1k} + \sum\limits_{i=n_1+1}^{n} \gamma_{ik}^{(t)}}$$

- **Stopping Criterion**:
  Let ϵ be a very small positive real number (e.g. 10^{-8}). The stopping (convergence) criterion is:

$$\left| \ell_{\text{obs}} \left(\boldsymbol{\theta}^{(t+1)}\right) \ell_{\text{obs}} \left(\boldsymbol{\theta}^{(t)}\right) \right| < \epsilon$$

  where

$$\ell_{\text{obs}}(\boldsymbol{\theta}) = \sum_{i=1}^{n_1} \log \left[\pi_{Z_i} f_{Z_i}(X_i)\right] + \sum_{i=n_1+1}^{n} \log \left[\sum_{k=1}^{2} \pi_k f_k(X_i)\right].$$

  is the observed-data log-likelihood